Metadata-Driven Secure Prompting (MDSP)

Data Analytics with GenAI and Metadata

Dr. Isaac K. Nti
Assistant Professor, School of Information Technology, University of Cincinnati, USA
Co-Lead Information Technology Analytics Center (ITAC)

What We Will Cover



Data analytics foundations
What data analytics is, its objectives, and common tools.



Three analytics approaches
Traditional, GenAI-assisted, and metadata-governed GenAI analytics.



Practical judgment
Strengths, limitations, risks, and when to use each approach.



Applied example
A simple institutional-data scenario.

What Is Data Analytics?

- **Data Analytics Turns Records into Evidence for Decisions**
  - **Raw records**
    - Survey responses, research measurements, grades, registrations, financial records, system logs, publications, and service requests.
  - **Analytical work**
    - Clean, organize, summarize, visualize, model, interpret, and validate.
  - **Decisions**
    - Improve teaching, research, institutional operations, policies, services, and strategy.
  - **Keep in Mind:**
    - Data analytics is not merely producing charts.
    - It is the disciplined process of converting data into defensible evidence.

What Can Analytics Help Us Do?

Describe

What happened?
Example: How many students enrolled in each program?

Diagnose

Why did it happen?
Example: Why did enrollment decrease?

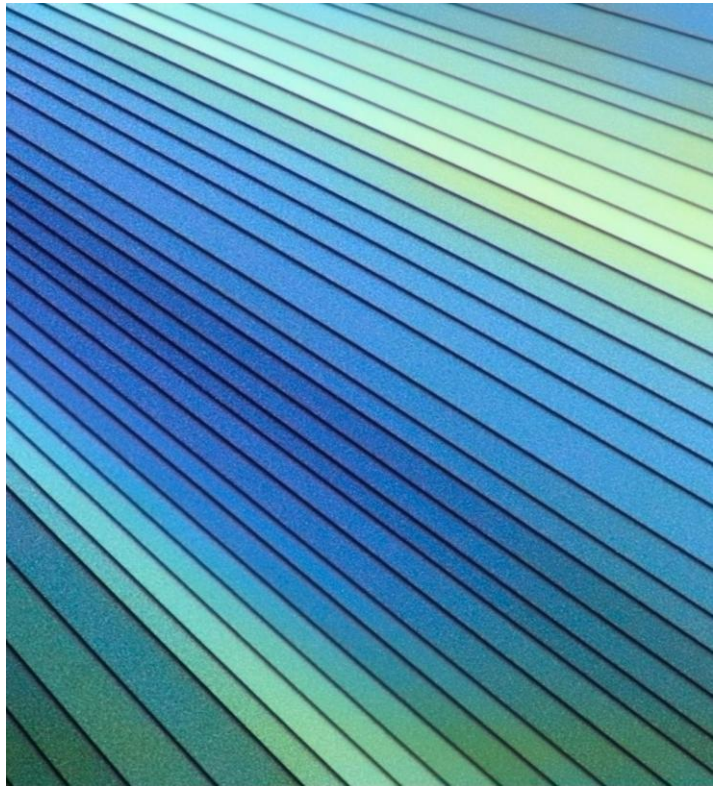
Predict

What may happen next?
Example: Which students may require additional academic support?

Prescribe

What should we do?
Example: Which intervention is most appropriate?

Analytics should move from information toward justified action.



Tools and Platforms

- Start with the Task, Not the Tool

Tool category	Examples	Suitable uses
Spreadsheets	Excel, Google Sheets	Cleaning, formulas, pivot tables, simple charts
Databases	SQL, PostgreSQL, BigQuery	Querying large structured datasets
Programming	Python, R, notebooks	Statistics, modeling, automation, reproducible analysis
Business intelligence	Power BI, Tableau	Dashboards, reporting, and monitoring
GenAI assistants	ChatGPT, Copilot, Gemini	Planning, code assistance, summaries, explanations
Metadata systems	Data dictionaries, catalogs	Definitions, ownership, quality, access, and lineage

Practical rule: Use GenAI to assist the analytical workflow, not as the sole calculator, evidence source, or auditor.

The 3 Non-Negotiables

- Before Beginning Any Analysis
  1. What decision or problem must the analysis address?
  2. May this data be used for this purpose and with this tool?
  3. How will we determine whether the result is sufficiently accurate, useful, and reliable?
- Without these three elements, even a visually impressive dashboard can be misleading.

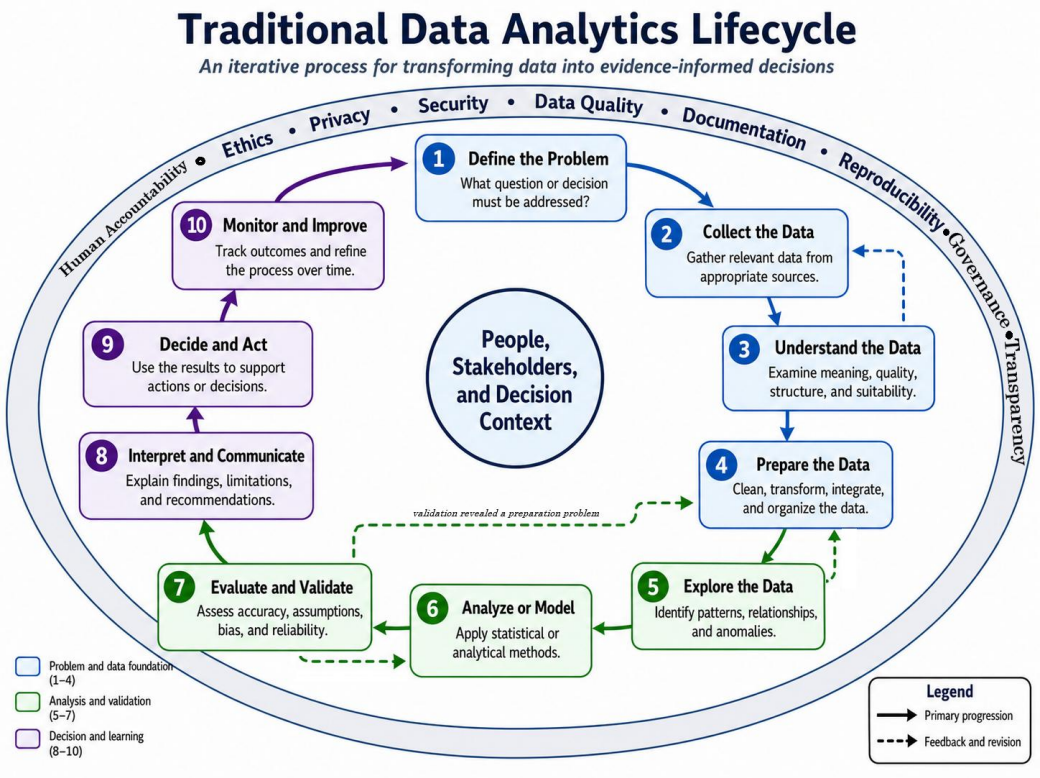
The 3 Data Analytics Approach

1. Traditional Data Analytics
2. GenAI-Assisted Data Analytics
3. Metadata-Governed GenAI Analytics (MDSP)

Traditional Data Analytics

Traditional analytics is not a waterfall; it loops when evidence requires correction.

- **Traditional analytics is a disciplined, iterative, and evidence-driven process.**
- **Main lifecycle**
  - Ask
  - Collect
  - Understand
  - Prepare
  - Explore
  - Analyze
  - Evaluate
  - Communicate
  - Act
  - Monitor
- **Important feedback loops**
  - Explore → Prepare
  - Evaluate → Analyze
  - Monitor → Ask again



Traditional Analytics: Strengths and Limitations

- **Strengths**
  - Strong reproducibility
  - Clear statistical reasoning
  - Easier auditing
  - Less dependence on prompts or AI-model behavior
  - Sensitive data can remain within approved local systems
  - Well-established methods and standards
- **Limitations**
  - May require substantial technical expertise
  - Exploration and documentation can be time-consuming
  - Data meaning may remain undocumented in people's knowledge
  - Communication may be slow
  - Unstructured text can require specialized methods
  - Repetitive tasks may require manual effort

Reliable, but can be slow without good documentation or automation.

GenAI-Assisted Data Analytics

(*What GenAI Changes*)

- GenAI Changes the Workflow, Not the Obligation to Verify
- **GenAI can assist with**
  - Analysis planning
  - SQL or code drafts
  - Formula generation
  - Chart recommendations
  - Text summarization
  - Plain-language explanations
  - Documentation
- **Analytical tools should perform**
  - Calculations
  - Query execution
  - Statistical tests
  - Model fitting
  - Chart rendering
  - Reproducible transformations

Humans remain responsible for

Purpose
Permission
Method selection
Validation
Interpretation
Decisions
Accountability

GenAI accelerates reasoning and communication; trusted analytical tools compute.

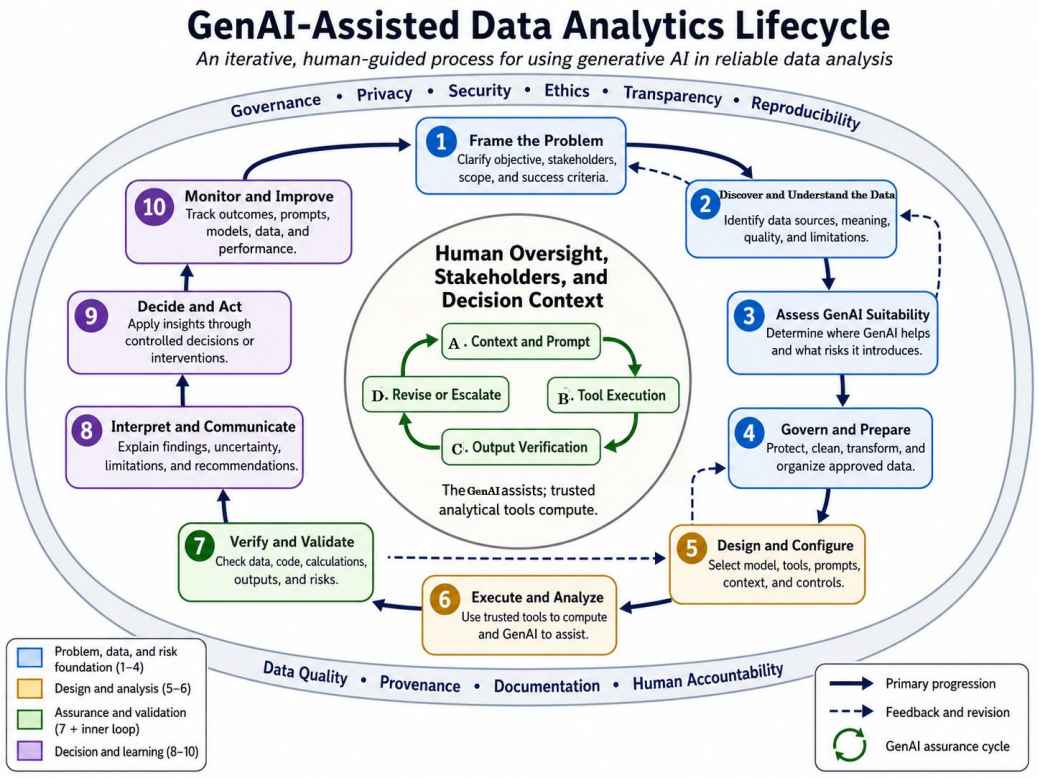
GenAI accelerates thinking and communication; tools still compute.

GenAI-Assisted Data Analytics

Same analytical backbone, plus a GenAI assurance loop.

Use GenAI inside a controlled cycle: design → prompt → execute → verify → revise.

- **Main lifecycle**
  - Frame the problem
  - Discover and understand the data
  - Assess GenAI suitability and risks
  - Govern and prepare the data
  - Design and configure the workflow
  - Execute and analyze
  - Verify and validate
  - Interpret and communicate
  - Decide and act
  - Monitor and improve
- GenAI-assurance cycle
  - Context and prompt → Tool execution → Output verification → Revise or escalate



Where GenAI Helps

The safest pattern is: ask AI to assist, then verify with data.

- *Use GenAI Where Language, Planning, and Coding Support Add Value*
  - **Plan**: Suggest analytical questions, methods, and workflow steps.
  - **Code**: Draft SQL, Python, R, spreadsheet formulas, or data-cleaning instructions.
  - **Explain**: Translate technical findings for students, administrators, researchers, or decision-makers.
  - **Summarize**: Condense reports, participant comments, literature, or open-ended survey responses.
  - **Visualize**: Recommend appropriate charts and communication narratives.
  - **Document**: Draft data dictionaries, analytical notes, and methodological descriptions.

Avoid: asking the GenAI to "just give the answer" when no computation or evidence has been executed.

GenAI-Assisted Data Analytics: Risk Controls

The question is not whether GenAI can help; it is whether the output is trustworthy.

Risk comes from unverified assistance.

Risk	Control
Hallucinated variables or records	Compare every reference against the actual schema
Incorrect code or formula	Execute, inspect, and test on known cases
Privacy exposure	Remove identifiers or provide metadata and aggregates only
Overconfident interpretation	Require evidence, uncertainty, and limitations
Inappropriate statistical method	Obtain analyst or domain-expert review
Model or version changes	Record model, prompt, date, settings, and tools
Fabricated sources	Open and verify every cited source

GenAI-Assisted Analytics: Strengths and Limitations

- **Strengths**
  - Faster analytical planning
  - Easier generation of code and formulas
  - Better plain-language explanations
  - Useful for text-heavy and document-based data
  - Reduces barriers for non-specialists
  - Speeds documentation and communication
  - Supports rapid exploration of alternatives
- **Limitations**
  - May hallucinate fields, values, or sources
  - Results can be prompt-sensitive
  - Privacy and data-sharing risks
  - May hide incorrect assumptions behind fluent language
  - Model behavior can change over time
  - Requires documented human verification
  - Can encourage overreliance or automation bias

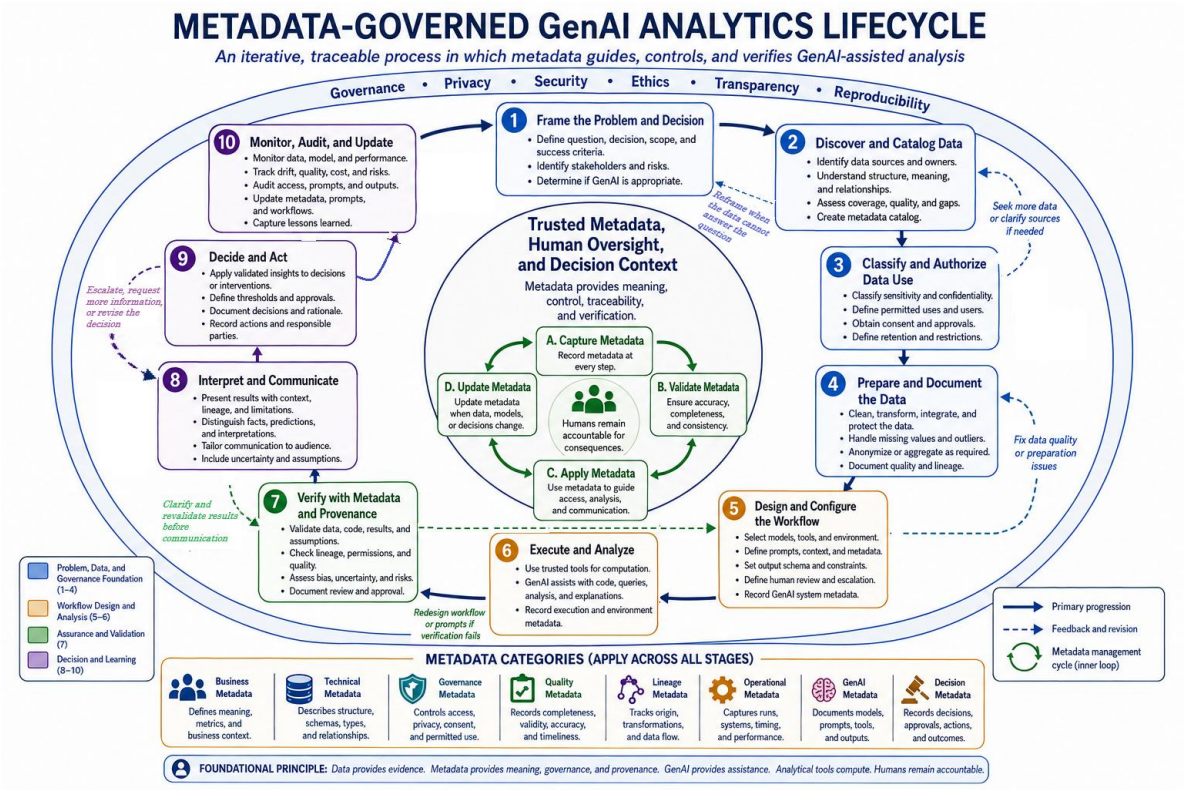
Metadata-Governed GenAI Analytics

Metadata-Driven Secure Prompting (MDSP)

- **Why Metadata Matters**
  - Metadata is information that describes the meaning, structure, ownership, quality, permissible use, and history of data.
- Without metadata
  - What does this variable mean?
  - Which unit is being used?
  - Can this field be shared?
  - Who owns the data?
  - Which dataset version is current?
  - Where did this result come from?
- With metadata
  - Definitions
  - Data types
  - Units
  - Sensitivity labels
  - Access restrictions
  - Quality indicators
  - Ownership
  - Lineage
  - Version history
- Metadata allows GenAI to operate with approved context and constraints.

Metadata-Governed GenAI Analytics: Steps

- Metadata becomes the context, control, provenance, and verification layer across the analytical lifecycle.
- **Main lifecycle**
  1. Frame the problem and decision
  2. Discover and catalog data and metadata
  3. Classify and authorize access and use
  4. Prepare and document data quality and lineage
  5. Design and configure models, tools, prompts, and controls
  6. 6Execute and analyze using trusted computation
  7. Verify using metadata and provenance
  8. Interpret and communicate with context and limitations
  9. Decide and act through controlled processes
  10. Monitor, audit, and update
- **Metadata-management cycle**
  - Capture → Validate → Apply → Update



Metadata Categories
(What Metadata Should We Capture?)

Metadata category	What it records
Business metadata	Definitions, indicators, rules, organizational meaning, research questions or objectives
Technical metadata	Schemas, data types, formats, and relationships
Governance metadata	Ownership, consent, access, sensitivity, and permitted use
Quality metadata	Completeness, validity, accuracy, and freshness
Lineage metadata	Sources, transformations, versions, and data flow
Operational metadata	Runs, costs, errors, timing, and system performance
GenAI metadata	Prompts, models, tools, parameters, and outputs
Decision metadata	Thresholds, approvals, actions, overrides, and outcomes

Metadata-Guided Prompting

- Give GenAI Approved Context Before Sensitive Data
- **Example metadata context**
- **Dataset**: Workshop registration data
- **Approved fields**
  - Country
  - Professional role
  - Area of interest
  - Attendance status
  - Consent status
- **Sensitive fields**
  - Full name
  - Email address
- **Permitted analysis**
  - Aggregate counts only
  - No identification of individuals
  - No inference of protected characteristics

Example prompt

Using only the approved metadata, propose three safe analyses for understanding participant composition. Do not expose individual records or infer personal characteristics. Provide an analysis plan and executable code, but do not invent results.

This separates analytical planning from unnecessary raw-data exposure.

Metadata-Governed Analytics: Strengths and Limitations

- **Strengths**
  - Better privacy and access control
  - Stronger traceability and auditability
  - More reliable prompts and generated code
  - Clearer field definitions and data interpretation
  - Easier collaboration across teams
  - Better reproducibility
  - Supports institutional memory and reuse
  - Stronger accountability for decisions
- **Limitations**
  - Metadata must be created and maintained
  - Initial setup takes time
  - Requires clear ownership and standards
  - May be excessive for very small one-time tasks
  - Requires governance knowledge
  - Metadata can become outdated
  - Requires organizational commitment and training

Comparison of the Three Approaches

There Is No Universal Best Approach

Criterion	Traditional analytics	GenAI-assisted analytics	Metadata-governed GenAI analytics
Typical speed	Moderate	Fast	Moderate after setup
Best suited for	Structured, reproducible analysis	Planning, coding, text, and explanation	Sensitive, shared, reusable, or institutional workflows
Main advantage	Established analytical rigor	Speed and accessibility	Context, governance, and traceability
Main risk	Slow or poorly documented work	Hallucination, privacy, and incorrect code	Metadata burden and governance overhead
Verification	Statistical and code checks	AI-output, code, evidence, and privacy checks	Metadata, lineage, access, code, and result checks
Accountability	Analyst or analytical team	Analyst and human reviewer	Data owner, analyst, reviewer, and decision-maker
Reusability	Moderate	Variable	High
Suitability for sensitive data	Good when kept local	Depends on platform and controls	Strongest when correctly implemented

Choosing the Right Approach

- **Match the Process to the Risk and Purpose**
  - **Low-risk, one-time task with familiar data**
    - Use traditional analytics.
  - **Need rapid planning, coding, summarization, or explanation**
    - Use GenAI-assisted analytics with verification.
  - **Sensitive, shared, regulated, institutional, or reusable workflow**
    - Use metadata-governed GenAI analytics.
  - **Permission is unclear or consequences are high**
    - Pause the analysis, resolve governance requirements, and require accountable human approval.
    - Use the lightest process that still protects validity, privacy, traceability, and accountability.

A Safe Prompt Pattern

- **Prompt for a Method, Not an Unsupported Answer**
- **Prompt template**
- **Role:** You are assisting with a data-analysis task.
- **Objective:** State the decision or analytical question.
- **Approved metadata:** Provide the schema, definitions, units, quality notes, and sensitivity classifications.
- **Allowed operations:** Specify whether the model may generate aggregate analysis, SQL, Python, visualizations, or summaries.
- **Restrictions:** Do not reveal identifiers, infer protected attributes, fabricate fields, or invent results.
- **Required output:**
  - Analysis plan
  - Recommended method
  - Executable code
  - Assumptions
  - Validation checks
  - Limitations
- Execute the code and compare every claim against the actual data.

Final Takeaways

- **Data provides evidence**
  - Claims must trace to records, documented sources, or executed calculations.
- **Metadata provides meaning and control**
  - Definitions, permissions, quality, ownership, and lineage guide appropriate use.
- **GenAI provides assistance**
  - It can support planning, code generation, explanation, summarization, and documentation.
- **Analytical tools compute**
  - Databases, spreadsheets, statistical software, and programming environments produce the actual calculations.
- **Humans remain accountable**
  - People verify the evidence, approve consequential uses, and make decisions.

FREE Data Source For Research

- Beginner-Friendly Dataset Sources
  - SynthDataHub: https://github.com/ntious/SynthDataHubLinks to an external site.
  - UCI Machine Learning Repository: https://archive.ics.uci.eduLinks to an external site.
  - Kaggle Datasets: https://www.kaggle.com/datasetsLinks to an external site.
  - Google Dataset Search: https://datasetsearch.research.google.comLinks to an external site.
  - FiveThirtyEight Datasets: https://data.fivethirtyeight.comLinks to an external site.
  - Data.world Dataset: https://data.worldLinks to an external site.
- Government and Public Data Sources
  - Data.gov: https://www.data.govLinks to an external site.
  - USA.gov Data and Statistics: https://www.usa.gov/statisticsLinks to an external site.
  - Data Cincinnati: https://data.cincinnati-oh.govLinks to an external site.
  - California Open Data Portal: https://data.ca.govLinks to an external site.
  - New York Open Data: https://opendata.cityofnewyork.usLinks to an external site.
  - U.S. Bureau of Labor Statistics: https://www.bls.govLinks to an external site.
  - Federal Reserve Economic Data (FRED): https://fred.stlouisfed.orgLinks to an external site.
- Science, Climate, and Space Data
  - NOAA Data Access: https://www.noaa.gov/dataLinks to an external site.
  - NASA Open Data Portal: https://data.nasa.govLinks to an external site.
- Research and Benchmark Dataset Sources
  - OpenML: https://www.openml.orgLinks to an external site.
  - Papers with Code Datasets: https://paperswithcode.com/datasetsLinks to an external site.
  - Hugging Face Datasets: https://huggingface.co/datasetsLinks to an external site.
  - GitHub Datasets: https://github.com/datasetsLinks to an external site.
  - Awesome Public Datasets: https://github.com/awesomedata/awesome-public-datasetsLinks to an external site.
- Finance, Business, and Economic Data
  - Yahoo Finance: https://finance.yahoo.comLinks to an external site.
  - World Bank Open Data: https://data.worldbank.orgLinks to an external site.
  - Quandl (Nasdaq Data Link): https://data.nasdaq.comLinks to an external site.
- Healthcare and Biomedical Data
  - PhysioNet: https://physionet.orgLinks to an external site.
  - NIH Data Resources: https://datascience.nih.gov/data-sharingLinks to an external site.
- Advanced and Large-Scale Data Sources
  - Registry of Open Data on AWS: https://registry.opendata.awsLinks to an external site.
  - Common Crawl: https://commoncrawl.orgLinks to an external site.
  - GDELT Project: https://www.gdeltproject.orgLinks to an external site.
  - RapidAPI: https://rapidapi.comLinks to an external site.
  - OpenWeather API: https://openweathermap.org/apiLinks to an external site.



Q&A

Hands-On

- Select one familiar dataset, document its essential metadata, and conduct a metadata-governed GenAI analysis.

Thank You

